

Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 08

AIM-Overcoming class imbalance using SMOTE techniques

Theory

- Describe different techniques for handling imbalance
- Working of **Synthetic Minority Oversampling Technique**(SMOTE) algorithm

Method

- 1) Analysis of the class label(target) to check if dataset is balanced or not
 - `Sns.countplot`
 - `data['Class'].value_counts()`
 - scatter plot of examples by class label
- 2)Now train the model without handling the imbalanced class distribution
- 3) Evaluate accuracy of model- Evaluation metric F-Score
- 4) Apply the algorithm SMOTE over sampling approaches
 - `SMOTE(sampling_strategy=0.1, k_neighbors=k)`. Try your model with different parameter values for sampling strategy and `k_neighbours`
- 5) Train the model after handling imbalance using SMOTE
- 5)Compare the evaluation results

Implementation Details

Dataset used

Library

```
imblearn.over_sampling import SMOTE
```